WEB APPLICATION DEVELOPMENT WITH NODE.JS

GV: TRẦN THANH TUẤN

07. BUILT-IN MODULES

File System Module

- The Node.js file system module allows you to work with the file system on your computer.

```
const fs = require('fs');
```

File System Module

- Common use for the File System module

Read files	•fs.readFile()
Create files	•fs.appendFile() •fs.open() •fs.writeFile()
Update files	•fs.appendFile() •fs.writeFile()
Delete files	•fs.unlink()
Rename files	•fs.rename()

File System Module

- Read files

```
fs.readFile('data.txt', 'utf-8', (err, data) => {
	if (err) {
		console.error(err);
		return;
	}
	console.log(data);
});
```

File System Module

- Create files

```
fs.appendFile('data.txt', 'Hello content!', function (err) {
	if (err) throw err;
	console.log('Saved!');
});
```

Appends specified content to a file. If the file does not exist, the file will be created

```
fs.open('data.txt', 'w', function (err, file) {
	if (err) throw err;

	fs.write(file, 'Hello content!', function (err) {
		if (err) throw err;
	});
	console.log('Saved!');
});
```

The flag is "w" for "writing", the specified file is opened for writing. If the file does not exist, an empty file is created

File System Module

- Create files

```
fs.writeFile('data.txt', 'Hello content!', function (err) {
        if (err) throw err;
        console.log('Saved!');
});
```

Replaces the specified file and content if it exists. If the file does not exist, a new file, containing the specified content, will be created.

File System Module

- Update files

```
fs.writeFile('data.txt', 'Hello content!', function (err) {
        if (err) throw err;
        console.log('Replaced!');
});
```

```
fs.appendFile('data.txt', 'Hello content!', function (err) {
        if (err) throw err;
        console.log('Updated!’);
});
```

File System Module

- Delete files

```
fs.unlink('data.txt', function (err) {
        if (err) throw err;
        console.log('File deleted!');
});
```

File System Module

- Rename files

```
fs.rename('data.txt', 'new-data.txt', function (err) {
	if (err) throw err;
	console.log('File renamed!');
});
```

HTTP Module

- Node.js has a built-in module called **HTTP**, which allows Node.js to transfer data over the Hyper Text Transfer Protocol (HTTP).
- The HTTP module can create an **HTTP server** that listens to server ports and gives a response back to the client.

```
const http = require('http');
```

HTTP Module

Represents the request from the client, as an object (http.IncomingMessage object)

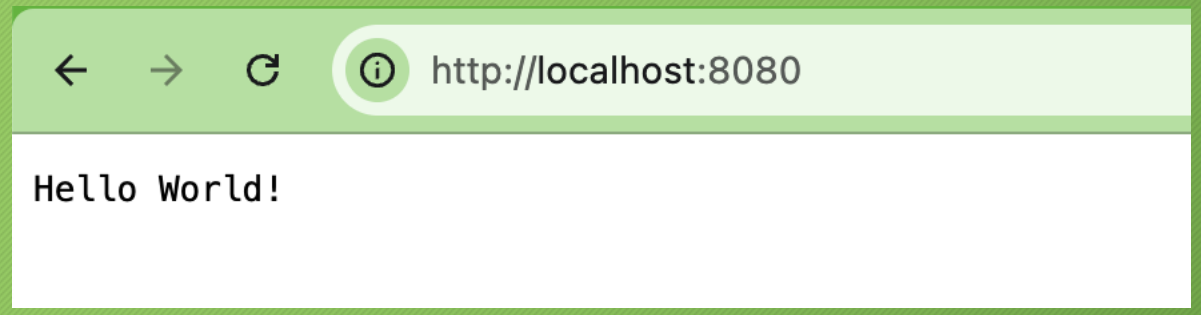
- Create a server object

```
http.createServer(function (req, res) {

    res.write('Hello World!');

    res.end();

}).listen(8080);
```

Write a response to the client

End the response

The server object listens on port 8080



HTTP Module

- Get url

```
req.url
```

- Add an HTTP Header

```
res.writeHead(200, {'Content-Type': 'text/html'});
```

URL Module

- The URL module splits up a web address into readable parts.

```
const url = require('url');
```

```
var adr = 'http://localhost:8080/default.htm?year=2017&month=february';
var q = url.parse(adr, true);

console.log(q.host);
console.log(q.pathname);
console.log(q.search);
var qdata = q.query;
console.log(qdata.month);
```

localhost:8080

/default.htm

?year=2017&month=february

{ year: 2017, month: 'february' }

february

Return a URL object with each part of the address as properties

Example 01

- Hello world

```
const http = require('http');

const server = http.createServer((req, res) => {
	res.writeHead(200, {'Content-Type': 'text/plain'});
	res.end('Hello World');
});

server.listen(3000, () => {
	console.log('Server đang chạy tại http://localhost:3000');
});
```

Example 02

- Hiển thị nội dung file HTML với File system module

```
const http = require('http');
const fs = require('fs');

const server = http.createServer((req, res) => {
	fs.readFile('index.html', 'utf-8', (err, data) => {
		if (err) {
			res.writeHead(404);
			res.end('File không tồn tại');
		} else {
			res.writeHead(200, {'Content-Type': 'text/html'});
			res.end(data);
		}
	});
});

server.listen(3000, () => {
	console.log('Server tại http://localhost:3000');
});
```

Example 03

- Xử lý query string với URL module

```
const http = require('http');
const url = require('url');

const server = http.createServer((req, res) => {
	const parsedUrl = url.parse(req.url, true);
	const name = parsedUrl.query.name || 'Stranger';

	res.writeHead(200, {'Content-Type': 'text/plain'});
	res.end(`Hello ${name}`);
});

server.listen(3000, () => {
	console.log('Truy cập http://localhost:3000/greet?name=Nam');
});
```

Example 04

- Đọc danh sách file trong thư mục

```
const http = require('http');
const fs = require('fs');
const url = require('url');

const server = http.createServer((req, res) => {
	const parsedUrl = url.parse(req.url, true);
	if (parsedUrl.pathname === '/files') {
		fs.readdir('./data', (err, files) => {
			if (err) {
				res.writeHead(500);
				res.end('Lỗi đọc thư mục');
			} else {
				res.writeHead(200, {'Content-Type': 'application/json'});
				res.end(JSON.stringify(files));
			}
		});
	} else {
		res.writeHead(404);
		res.end('Not Found');
	}
});

server.listen(3000, () => {
	console.log('Truy cập http://localhost:3000/files');
});
```

Example 05

- Ghi nội dung vào file từ query string

```
const http = require('http');
const url = require('url');
const fs = require('fs');
```

```
const server = http.createServer((req, res) => {
	const parsed = url.parse(req.url, true);
	if (parsed.pathname === '/write') {
		const fileName = parsed.query.file;
		const content = parsed.query.content;

		if (!fileName || !content) {
			res.writeHead(400);
			res.end('Thiếu file hoặc nội dung');
		} else {
			fs.writeFile(`./data/${fileName}`, content, (err) => {
				if (err) {
					res.writeHead(500);
					res.end('Lỗi ghi file');
				} else {
					res.writeHead(200);
					res.end(`Đã ghi vào ${fileName}`);
				}
			});
		}
	} else {
		res.writeHead(404);
		res.end('Not Found');
	}
});
```

Example 05

- Ghi nội dung vào file từ query string

```
server.listen(3000, () => {
	console.log('Truy cập ví dụ: http://localhost:3000/write?file=test.txt&content=Hello');
});
```

Exercises 01

- Viết server trả về nội dung khác nhau tùy theo đường dẫn:
  - / → Trang chính
  - /about → Giới thiệu
  - /contact → Liên hệ
- Nội dung các trang được load từ các file html tương ứng
  - index.html
  - about.html
  - contact.html

Exercises 02

- Ghi log truy cập vào file log.txt
  - Mỗi lần có truy cập, server ghi log thời gian và URL vào log.txt

Thank you!